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Archives of Microbiology

ISSN 0302-8933

Arch Microbiol

DOI 10.1007/s00203-020-02021-8



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Antibiotic-resistant Enterobacteriaceae from diseased freshwater goldfish

P. G. Preena¹ · Arathi Dharmaratnam¹ · N. Sundar Raj¹ · S. Arun Raja¹ · Reshma R. Nair¹ · T. Raja Swaminathan¹

Received: 3 June 2020 / Revised: 3 August 2020 / Accepted: 10 August 2020
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Abstract

Goldfish farming gained more attention among the ornamental fishes in aquaculture industry. The occurrence of bacterial infections and further antimicrobial treatment lead to the major crisis of antibiotic resistance in aquaculture. We have isolated diverse enterobacteriaceae groups which affect the goldfish and identified their response towards 46 antimicrobials of 15 different classes. Thirteen significant bacterial isolates such as *Edwardsiella tarda*, *Serratia marcescens*, *Klebsiella aerogenes*, *Proteus penneri*, *P. hauseri*, *Enterobacter cloacae*, *E. cancerogenus*, *E. ludwigii*, *Citrobacter freundii*, *E. coli*, *Kluyvera cryocrescens*, *Plesiomonas shigelloides* and *Providencia vermicola* were recovered from the infected fish with the Shannon–wiener diversity index of 2.556. Multiple antibiotic resistance (MAR) index was found to be maximum for *P. penneri* (0.87) and minimum for *C. freundii* and *E. cloacae* (0.22), highlighting the hyper antibiotic selection pressure in the farm. The minimum concentration of antibiotics required to inhibit most of the resistant isolates was found to be > 256 mcg/ml. All the isolates were susceptible towards ciprofloxacin. Plasmid curing and further AMR tests could reveal the location of antibiotic resistance genes mainly as plasmids which determine the large extent of AMR spread through horizontal gene transfer. This study is the first of its kind to investigate the antimicrobial resistance profile of enterobacteriaceae recovered from goldfish, before and after plasmid curing.

Keywords Antibigram profile · Antimicrobial resistance · Enterobacteriaceae · Goldfish · Plasmid curing

Introduction

Nowadays, there is a global increase in ornamental fish farming in the aquaculture industry. Singapore, Japan, Czech Republic, Germany and Malaysia are some of the prime countries in exporting ornamental fishes, while above 50% of aquarium fishes are globally exported by Asian countries (Mohammad et al. 2018). Goldfish is one of the most popularly cultured ornamental fishes in India (FAO 2005) and gained more attention in the pet industry as a large number

of varieties are available (Mirzaei and Khovand 2015). In the earlier period onwards a mass mortality of goldfish has been reported due to severe bacterial infections (Dixon and Issvoran 1993). Gram-negative bacteria are the major pathogens affecting ornamental fishes (Carnevia et al. 2013). Several pathogens belonging to enterobacteriaceae groups viz *Proteus*, *Citrobacter*, *Edwardsiella*, *Enterobacter*, *Plesiomonas*, and *E. coli* are reported to induce severe infections in ornamental fishes (Walczak et al. 2017). Antimicrobials such as tetracyclines, beta lactams, quinolones, nitrofurans and aminoglycosides are commonly used to control bacterial infections in the ornamental fish industry (Verner-Jeffreys et al. 2009). Antibiotics are routinely applied to the aquaculture system via feed supplements or immersion baths to suppress the growth of potential pathogens and also as growth promoters (Ibrahim et al. 2010). The excess antibiotics remain unaltered and could persist in the cultured system for long periods, which leads to the development of selection pressure for antimicrobial resistance (AMR) (Goni-Urriza et al. 2000). Severe levels of antibiotic resistance towards β lactam antibiotics were observed in enterobacteriaceae

Communicated by Erko Stackebrandt.

Electronic supplementary material The online version of this article (<https://doi.org/10.1007/s00203-020-02021-8>) contains supplementary material, which is available to authorized users.

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groups such as *Klebsiella*, *Proteus*, *Serratia*, *Comamonas* and *Edwardsiella* (Shaikh et al. 2015). This could affect the successful recovery of fishes infected with such potential pathogens. In addition, the transmission of such zoonotic AMR bacteria to humans during handling poses public health risks (Sharma et al. 2006). Hence the occurrence of antibiotic resistance in such pathogens needs to be monitored at regular intervals to overcome the AMR menace.

The mobile genetic elements (MGE) such as plasmids and transposons integrated with integrons/gene cassettes are one of the important mediators in spreading antibiotic resistance (Preena et al. 2020a). The antibiotic-resistant genes (ARG) can be exchanged among the fish pathogens through horizontal gene transfer (HGT) which could further promote the emergence of genetic mutations in chromosomal DNA (Strahilevitz et al. 2009). Hence the role of pathogens with such MGE in acting as a potential reservoir for antibiotic resistance should be routinely assessed in aquaculture systems (Verner-Jeffreys et al. 2009). However, various studies reported the existence of antibiotic-resistant genes in bacterial chromosome also (Amalina et al. 2019). This information reminds the transmission of AMR to next generation even through vertical gene transfer other than HGT. The location of ARG, whether plasmid/chromosome, in bacterial strains could be detected through plasmid curing assay (Letchumanan et al. 2015a). Plasmid curing is the method of eliminating plasmids by means of effective chemical agents such as ethidium bromide, acridine orange and sodium dodecyl sulphate (Liu et al. 2012). Acridine orange was found to be the best agent for plasmid curing (Molina-Aja et al. 2002) and hence it has been utilized in our study. Since AMR genes located within the plasmid got eliminated during plasmid curing, further antimicrobial susceptibility tests may determine the location of ARG. Plasmid profiling of antimicrobial-resistant aeromonadaceae and vibrionaceae family members from different cultured fishes have already been performed (Giles et al. 1995; Amalina et al. 2019). However, plasmid curing studies of enterobacteriaceae groups associated with goldfish were hardly studied so far. Understanding the mode of antimicrobial resistance dissemination is crucial in implementing control strategies. Thus, the present study was initiated to evaluate the antibiotic resistance patterns of different species of enterobacteriaceae groups recovered from the diseased goldfish, *Carassius auratus* before and after plasmid curing.

Materials and methods

Sample collection and processing

Moribund goldfish (*Carassius auratus*) ($n=140$) were collected from ornamental fish farms situated in Kerala and

Tamil Nadu, during the period between April 2018 and January 2020. A total of seven sets of sampling had done and twenty diseased fish were collected during each sampling. The freshly dead and moribund fish were collected on ice and transported to lab. The clinical signs observed in each fish were recorded.

Isolation of bacteria belonging to enterobacteriaceae family from goldfish

The fish were processed aseptically for the collection of gills and internal organs such as liver, kidney and spleen. The same type of tissues from the diseased twenty fish, of each sampling, were homogenized and pooled together and then inoculated in nutrient broth and incubated overnight at 28 °C. For the isolation of enterobacteriaceae groups, the serially diluted cultures were inoculated on MacConkey media (Himedia, India) using conventional spread plate technique. The resolved colonies were selected based on the unique morphology and sub cultured on MacConkey agar plates and stored in glycerol at -80°C for further studies. The isolates had then undergone biochemical tests such as gram staining, motility tests, catalase, oxidase, IMViC, nitrate reduction and carbohydrate fermentation tests, for the preliminary identification of enterobacteriaceae groups (Cowan and Steel 1974).

Diversity determination using Primer-E software and NtSys software

The isolates which showed biochemical characteristics similar to that of enterobacteriaceae were selected for the present study. On the basis of a number of isolates of each type of enterobacteriaceae recovered from the diseased fish, Shannon–Wiener diversity index has been determined using Primer E software (Clarke and Gorley 2015). To avoid the sequencing of all the separated isolates, dendrogram has been drawn based on the phenotypic characters using NtSys software. For this, positive and negative results of biochemical tests were documented on excel sheets and then transferred to NTedit version 1.1b and using NTSyspc, version 2.02i (Exeter Software, Setauket, NY, USA). Thus, the representative isolates clustered under each branch of dendrogram were subjected to further studies.

DNA extraction and 16SrRNA gene sequencing and phylogenetic characterisation

Total genomic DNA of the representative isolates has been extracted by salting out method following Miller et al. (1988). Quantity and quality of DNA were checked at the absorbance of 260 nm and 280 nm on UV spectrophotometer (Beckman, USA). PCR amplification was performed for

16SrRNA gene using the primers fD1 (5'CCG AAT TCG TCG ACA ACA GAG TTT GAT CCT GGC TCA G3') and rD1 (5'CCC GGG ATC CAA GCT TAA GGA GGT GAT CCA GCC3') (Weisburg et al. 1991). The PCR products were purified and sequenced using ABI Prism 3700 Big Dye sequencer (SciGenom, Cochin). Using NCBI BLAST, the gene sequences were identified by comparing with the sequences deposited in Genbank databases. Phylogenetic analysis was carried out using UPGMA statistical method and Kimura 2-parameter substitution model employing Mega 10 version (Tamura et al. 2011).

Antimicrobial resistance profiling

The identified isolates were subjected to antibiotic susceptibility tests using disc diffusion method on Mueller–Hinton agar (HiMedia, India) (CLSI 2018). As given in Table 1, forty-six antibiotics belonging to different classes were used to determine the antibiotic resistance pattern. The antibiotic discs (HiMedia, India) were placed on the pure culture inoculated plates and incubated overnight at 28 °C. The size of the inhibition zone (mm) was measured and the results were recorded according to CLSI (2018). The multiple antibiotic resistance (MAR) index, ratio of resistant antibiotics to a total number of tested antibiotics, of the tested isolates was calculated as per the antibiotic susceptibility test results (Krumperman 1983).

Minimum inhibitory concentration of the resistant isolates

The minimum inhibitory concentration (MIC) of antibiotics was calculated for the resistant isolates using MIC strips (Himedia, India) overlaid with different concentration of each antibiotic. All the test cultures were kept at the same turbidity (10^5 CFU/ml) on comparing with the standard, 0.5 McFarland solution. The uniformly maintained cultures were spread on Muller Hinton agar plates and three MIC strips/plate were kept on the inoculated plates. A positive control, CLSI reference strain *E. coli* ATCC 25922 was also inoculated. All the plates were incubated at 28 °C for 24 h. The antibiogram profile has been elucidated according to CLSI (Krumperman 1983).

Plasmid extraction of antimicrobial-resistant isolates

A total of 2 ml of overnight grown antimicrobial-resistant cultures were centrifuged at 12,000g for 2 min at 37 °C. The plasmids of pelletized samples were extracted using a plasmid mini preparation kit (Sigma Aldrich, USA). The eluted plasmids were visualized using 1% agarose gel

Table 1 Details showing the percentage of antibiotic-resistant goldfish isolates towards 46 antibiotics

Antimicrobials	Disc content (µg)	% of resistant isolates from Goldfish
β-Lactams		
Amoxicillin (AMX)	25	84.6
Amoxycylav (AMC)	30	38.4
Ampicillin (AMP)	25	100
Aztreonam (AT)	30	61.5
Piperacillin (PI)	100	38.4
Piperacillin/tazobactam (PIT)	100	38.4
First-generation cephalosporins		
Cefalexin (CN)	30	46.1
Cefazolin (CZ)	30	69.2
Cefuroxime (CXM)	30	46.1
Cephalothin (CEP)	30	76.9
Second-generation cephalosporins		
Cefoxitin (CX)	30	30.7
Third-generation cephalosporins		
Cefixime/clavulanic acid (CMC)	5	38.4
Cefoperazone (CPZ)	75	46.1
Cefotaxime (CTX)	30	38.4
Cefotaxime/Clavulanic acid (CEC)	30	15.3
Ceftazidime (CAZ)	30	53.8
Ceftazidime/clavulanic acid (CAC)	30	38.4
Ceftriaxone (CTR)	30	7
Fourth-generation cephalosporins		
Cefepime (CPM)	30	38.4
Carbapenem		
Doripenem (DOR)	10	15.3
Imipenem (IPM)	10	15.3
Meropenem (MRP)	10	23
Aminoglycosides		
Amikacin (AK)	30	38.4
Gentamycin (HLG)	30	46.1
Streptomycin (S)	10	46.1
Tobramycin (TOB)	10	38.4
Macrolides		
Azithromycin (AZM)	30	69.2
Erythromycin (E)	15	69.2
Quinolones and fluoroquinolones		
Ciprofloxacin (CIP)	30	0
Enrofloxacin (EX)	10	15.3
Levofloxacin (LE)	5	0
Norfloxacin (NX)	10	23
Nalidixic acid (NA)	30	61.5
Perfloxacin (PF)	5	0
Tetracyclines		
Oxytetracycline (O)	30	76.9
Sulphonamides		
Co-trimoxazole (COT)	25	23

Table 1 (continued)

Antimicrobials	Disc content (µg)	% of resistant isolates from Goldfish
Sulphadiazine (SZ)	100	53.8
Trimethoprim (TR)	25	15.3
Phenicol		
Chloramphenicol (C)	30	23
Nitrofurans		
Furazolidone (FR)	100	38.4
Nitrofurantoin (NIT)	100	69.2
Glycopeptides		
Bacitracin (B)	10	84.6
Vancomycin (VA)	30	100
Other antibiotics		
Colistin (CL)	10	61.5
Polymyxin-B (PB)	300	46.1
Rifampicin (RIF)	5	69.2

electrophoresis and documented in a gel documentation system (Bio-Rad Laboratories, California, USA).

Plasmid curing and antibiogram profiling

The isolates which harbored plasmids were subjected to plasmid curing using acridine orange (AO) (Sigma Aldrich, USA) following Letchumanan et al. (2015b). A loopful of antibiotic-resistant bacterial cultures was grown individually on nutrient broth supplemented with different range (0.1–1 mg/ml) of AO and placed in shaker incubator overnight at room temperature. To verify the changes in the antibiogram profile after the AO treatment, the antibiotic susceptibility tests were repeated for the resistant isolates as described previously. MAR index of the resistant isolates was again calculated after plasmid curing.

Results and discussion

Isolation of enterobacteriaceae groups and diversity determination

Most of the infected fish exhibited abnormal swimming, lethargy, anorexia and hemorrhages and some have shown discoloration and gill necrosis. On post mortem, kidney and liver congestion, spleen enlargement and ascitic fluid accumulation were noticed. Numerous colonies with different morphologies were obtained from the pooled organs of each set of fish samples on spread plating. Altogether 84 suspected diverse enterobacteriaceae isolates have been selected based on the unique characteristics of MacConkey agar. Biochemical characteristics of recovered isolates for

selecting the representative organisms are given in Table 2. Shannon–weaver diversity has been traditionally used to determine the microbial communities and it is directly proportional to species richness and evenness (Kim et al. 2017). The diversity index of the isolates recovered from the goldfish has been calculated as $H' (\log_e) = 2.556$ using Primer E software. This is in turn influenced by the diversity factors such as species richness ($d = 2.708$), species evenness ($J' = 0.9964$) and species dominance ($(1 - \lambda) = 0.9329$). Out of the 84 isolates tested, 13 clusters were formed on the dendrogram (Supplementary ESM Fig. 1) based on biochemical test results using NtSys. This indicated that the recovered isolates are of 13 types.

The sequences of PCR amplified 16SrRNA gene products of 13 representative isolates had shown significant similarity to that of type strains deposited in Genbank database collections. The sequences had exhibited 99.86% similarity to *Edwardsiella tarda*, 99.1% to *Citrobacter freundii*, 99.71% to *Klebsiella aerogenes*, 99.43% to *Enterobacter cancerogenus*, 98.27% to *Enterobacter cloacae*, 98.94% to *Enterobacter ludwigii*, 99.73% to *Proteus hauseri*, 99.85% to *Proteus penneri*, 99.72% to *Plesiomonas shigelloides*, 99.59% to *Providencia vermicola*, 99.66% to *Kluyvera cryocrescens* and 99.73% to *Serratia marcescens* upon comparison with the corresponding type strains. The identified 16SrRNA gene sequences were submitted to Genbank with accession numbers MT263016–MT263028. The phylogenetic tree (Fig. 1) constructed using Mega 10 software revealed a dendrogram containing 13 operational taxonomic units (OTUs) along with the corresponding type strains from Genbank database. Since all the organisms belong to the enterobacteriaceae family, the distance coefficient of the phylogenetic tree is very low, 0.030 to 0.00. The tree mainly consisted of two branches separated at the distance coefficient of 0.035r. *P. vermicola*, *P. hauseri* and *P. penneri* were grouped under the same branch and separated from other groups at 0.035r. Here, *P. hauseri* and *P. penneri* were clustered together by separating from *P. vermicola* at 0.025r. While the remaining 10 groups were clustered under the other branch. *Enterobacter* groups such as *E. cancerogenus*, *E. cloacae* and *E. ludwigii* being closely related and belong to the same genus, were clustered together. *C. freundii*, *K. aerogenes* and *K. cryocrescens* were branched to *Enterobacter* groups at less than 0.01r similarity coefficient, indicating their close relatedness. *S. marcescens*, *E. coli*, *E. tarda* and *P. shigelloides* were then further joined to the above groups at 0.015r, 0.020r, 0.025r and 0.030r, respectively. The overall mean distance between the 16SrRNA gene sequences of isolates as calculated using Mega 10 software is 0.05.

In the present study, *E. tarda* and *P. hauseri* were recovered from the infected organs such as liver and kidney of all the sets of fish processed. While *S. marcescens* has been identified among the gills and skin surfaces of four sets of

Table 2 Biochemical test results of recovered goldfish isolates

Species	No. of isolates recovered from gold fish	Gram stain	Motility	Catalase	Oxidase	Indole	Methyl red	Voges proskauer	Citrate utilization	Nitrate reduction tests	Carbohydrate fermentation test		
											Glucose	Sucrose	Lactose
<i>Klebsiella aerogenes</i>	6	Gram -ve	+ve	+ve	-ve	-ve	-ve	+ve	+ve	+ve	Acid/gas	Acid/gas	Acid/gas
<i>Kluyvera cryocrescens</i>	6	Gram -ve	+ve	+ve	-ve	+ve	+ve	-ve	+ve	+ve	Acid/gas	Acid/gas	Acid/gas
<i>Proteus hauseri</i>	7	Gram -ve	+ve	+ve	-ve	-ve	+ve	-ve	+ve	+ve	Acid/gas	Acid/gas	-ve
<i>Proteus permeri</i>	7	Gram -ve	+ve	+ve	-ve	-ve	ve	-ve	-ve	+ve	Acid/gas	Acid/gas	-ve
<i>Citrobacter freundii</i>	7	Gram -ve	+ve	+ve	-ve	-ve	+ve	-ve	+ve	+ve	Acid/gas	Acid/gas	Acid/gas
<i>E. coli</i>	7	Gram -ve	+ve	+ve	-ve	+ve	+ve	-ve	-ve	+ve	Acid/gas	Acid/gas	Acid/gas
<i>Plesiomonas shigelloides</i>	6	Gram -ve	+ve	+ve	+ve	+ve	+ve	-ve	-ve	+ve	Acid/gas	Acid/gas	Acid/gas
<i>Enterobacter cloacae</i>	7	Gram -ve	+ve	+ve	-ve	-ve	-ve	+ve	+ve	+ve	Acid/gas	-ve	Acid/gas
<i>Enterobacter cancerogenus</i>	7	Gram -ve	+ve	+ve	-ve	-ve	-ve	+ve	+ve	+ve	Acid/gas	Acid/gas	-ve
<i>Enterobacter ludwigii</i>	7	Gram -ve	+ve	+ve	-ve	-ve	-ve	+ve	+ve	+ve	Acid/gas	-ve	-ve
<i>Edwardsiella tarda</i>	7	Gram -ve	+ve	+ve	-ve	+ve	+ve	-ve	-ve	+ve	Acid/gas	Acid/gas	-ve
<i>Serratia marcescens</i>	4	Gram -ve	+ve	+ve	-ve	-ve	-ve	+ve	+ve	+ve	Acid/gas	Acid/gas	-ve
<i>Providencia vermicola</i>	7	Gram -ve	+ve	+ve	-ve	+ve	+ve	+ve	+ve	+ve	Acid/gas	-ve	Acid/gas

tested samples. It is reported that goldfish has been used as a suitable infection model for studying the immune response against *Edwardsiella* infection (Choe et al. 2017). Virulent *S. marcescens* was recently reported in a disease outbreak occurred in an ornamental fish farm and characterized by Dharmaratnam et al. (2017). Kumar et al. (2015) reported higher mortality rates among the ornamental fish due to antibiotic resistant *P. hauseri* infection. All these reports indicate the severity of infection associated with the studied fish. *Enterobacter cancerogenus*, *Enterobacter cloacae* and *Enterobacter ludwigii* were recovered from every infected organ of the tested fish. The human enteric bacteria like *E. cloacae* could induce diseases which lead to heavy mortality in fishes (Sekar et al. 2008). Hence the recovery of all three species of *Enterobacter* in the present study could raise challenges. Besides, *C. freundii* and *E. coli* were also segregated from the internal parts, such as liver and spleen, respectively, of all the sample sets; while *K. aerogenes*, *K. cryocrescens*, *P. vermicola* and *P. shigelloides* from the organs of six samples sets only. Similar to this study, *Escherichia*, *Citrobacter*, *Klebsiella*, *Enterobacter*, and *Proteus* were found to associate with the cultured fishes (Gufe et al. 2019). They pointed out the source of enteric bacteria as fecal contamination and which pose health risks to human and animals. The fecal matter of birds and animals in the farm might be the source of such enteropathogens in the current study also. *Providencia vermicola* was found to infect carp by showing reddish coloration and ulcers on abdomen and which was the first report of *Providencia* in causing disease in cultured fishes (Ramkumar et al. 2014). Likewise, *Plesiomonas shigelloides* also exhibited severe pathogenicity among the cultured carps (Behera et al. 2018). The representatives of enterobacteriaceae which are associated with fish diseases and reports of great losses in the aquaculture industry have been reviewed in detail by Austin and Austin (2016). Thus, the identification of these pathogens belonging to enterobacteriaceae enforced us to perform antimicrobial susceptibility studies for a better understanding of their response to diverse antibiotics. The studies on the existence of antibiotic resistance in fish pathogens and the way of AMR dissemination in aquaculture are crucial.

Antibiogram profile

Antimicrobial susceptibility of the 13 representative enterobacteriaceae isolates revealed that they were resistant to a minimum of 10 and a maximum of 40 antibiotics tested. The details showing the percentage of isolates showing antimicrobial resistance towards 46 antibiotics and fifteen different antibiotic classes are given in Table 1 and Fig. 2 respectively. Thirty-eight percent of the isolates were resistant to more than 25 antibiotics and remaining sixty-two percent were resistant towards more than 10 antibiotics. Out of the 15

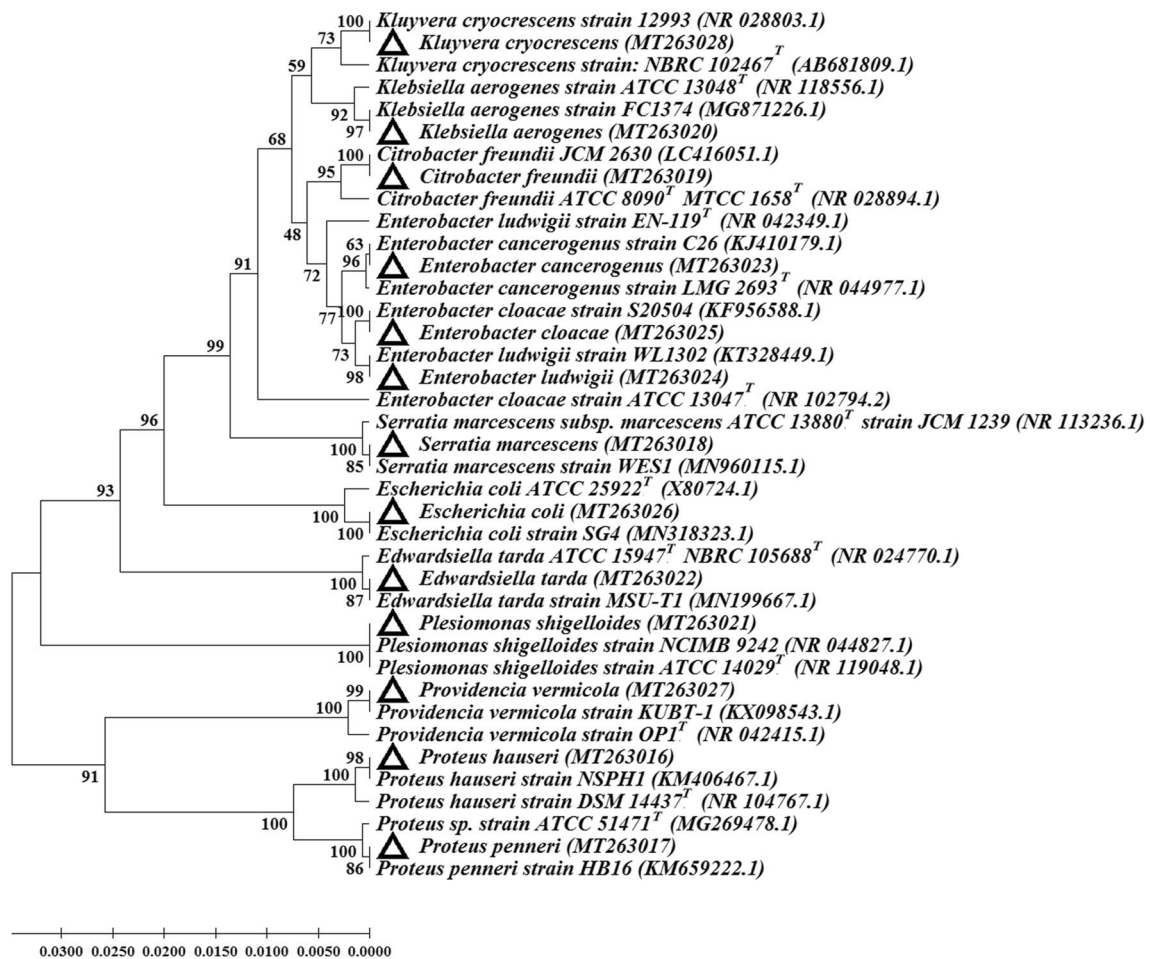
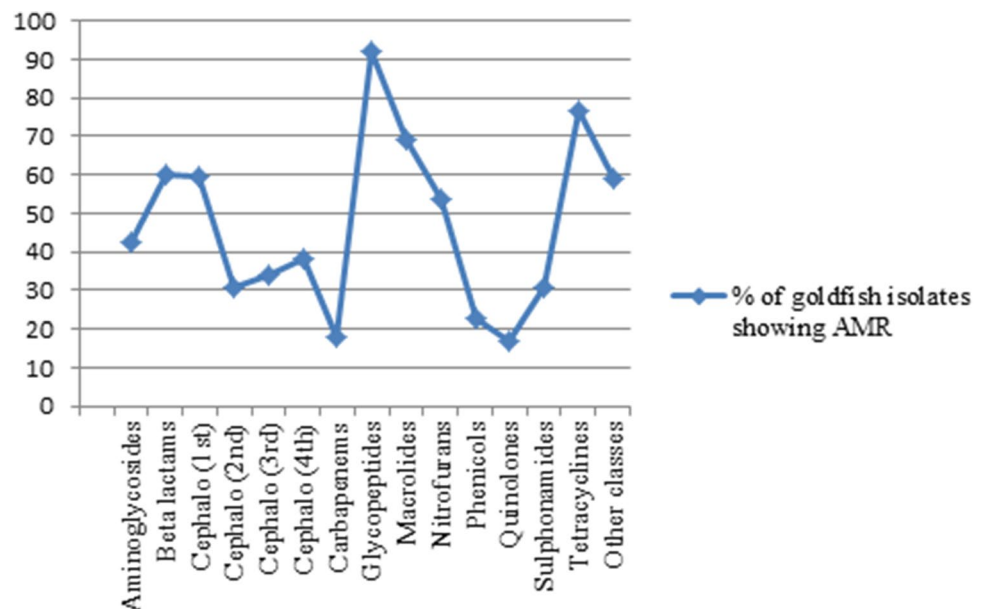


Fig. 1 Phylogenetic analysis of the 16S rRNA gene sequences of goldfish isolates using Mega10 software. The UPGMA tree constructed was computed using Kimura-2 parameter substitution model with 1000 bootstrap replications

Fig. 2 A graph showing the percentage of goldfish isolates showing AMR towards 15 classes of antibiotics

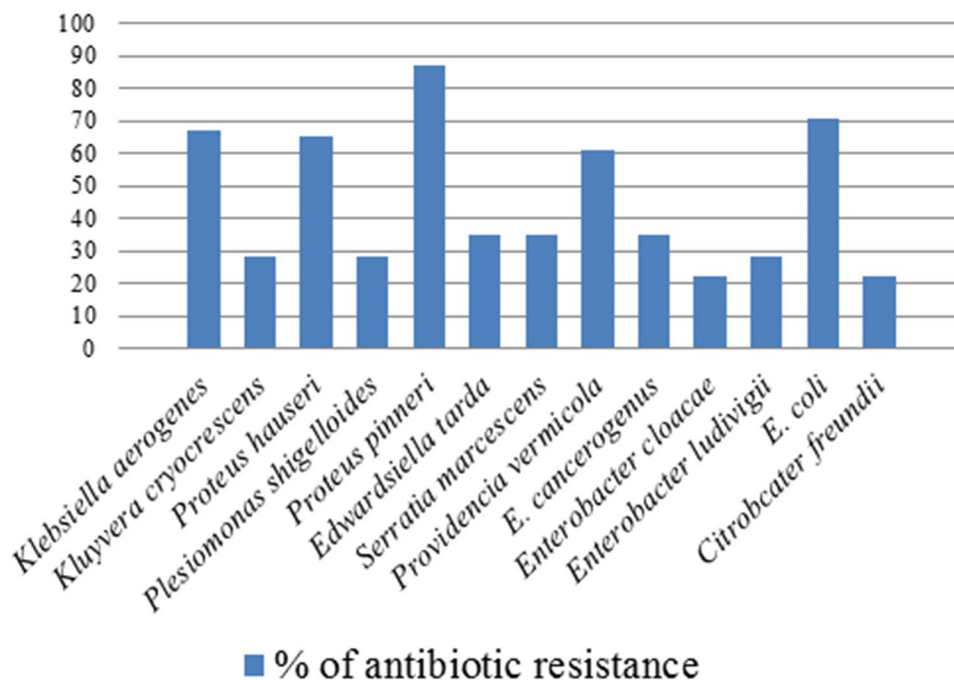


classes of antibiotics tested, a minimum of at least one antibiotic of 7 classes and a maximum of that of 15 classes were shown to be resistant by the studied isolates. This indicates the severity of the occurrence of antibiotic resistance in the aquaculture farm under investigation. The graph showing the percentage of antibiotic resistance of recovered isolates towards the 46 antibiotics tested is depicted in Fig. 3. The minimum inhibitory concentration of antibiotics for most of the resistant isolates, as detected using MIC strips (Himedia), was found to be very high (> 256 mcg/ml). The MIC of tested antibiotics for all the resistant recovered isolates is given in supplementary ESM Table 1. *P. penneri* was found to be the top most multi drug resistant isolate compared to the other ones with the higher MAR index of 0.87. It is to be pointed out that *P. penneri* was resistant towards 40 antibiotics, including new generation cephalosporins and carbapenems, belonging to fifteen classes. Next to *P. penneri*, other major isolates such as *E. coli*, *K. aerogenes*, *P. hauseri* and *P. vermicola* exhibited higher antibiotic resistance with the MAR index of 0.71, 0.67, 0.65 and 0.61, respectively. Here, *E. coli* exhibited resistance towards 33 antibiotics of 12 classes, and *K. aerogenes* against 31 antibiotics of 13 classes. While *P. hauseri* and *P. vermicola* showed resistance towards 30 antibiotics of 14 classes and 28 of 13 classes, respectively. Fourth-generation cephalosporin, cefepime was found to be ineffective against the resistant isolates such as *P. penneri*, *K. aerogenes*, *P. hauseri*, *P. vermicola* and *P. shigelloides*. Likewise the most effective carbapenem groups were observed as ineffective for *K. aerogenes*, *P. penneri*, *E. coli* and *E. cancerogenus*. This again denotes the gravity of the antibiotic resistance development in the farm.

Another significant isolate, *S. marcescens* also conferred multiple antibiotic resistance towards 16 antibiotics, including third-generation cephalosporins, belonging to 10 different classes with the MAR index of 0.35. In addition to carbapenem groups, *E. cancerogenus* showed AMR towards 16 antibiotics of 9 classes (MAR index = 0.35). Similarly, *E. tarda* also exhibited the same multiple antibiotic resistance index of 0.35 by showing resistance towards 16 antibiotics of 9 classes. However, *K. cryocrescens*, *P. shigelloides* and *E. ludwigii* are comparatively susceptible isolates by showing resistance only against 13 antibiotics of 7 classes, with the MAR index of 0.28. This is followed by *C. freundii* and *E. cloacae*, which are found to be susceptible towards 37 tested antibiotics and showed resistance towards only 10 antibiotics of 7 classes with the MAR index of 0.22. Similar to the present observation, Preena et al. (2019b) had also successfully isolated multiple drug-resistant *K. aerogenes*, *E. cancerogenus*, *E. cloacae*, *E. tarda* and *C. freundii*, with higher MAR index, from goldfish and koi carp. A MAR index of greater than 0.2 indicates the severe usage of antibiotics in aquaculture or the heavy entry of antibiotics from some other sources (Krumperman 1983). In the present study, all of the isolates possessed MAR index > 0.2 with a range of 0.21 to 0.85 and mean value of 0.45. This highlights the heavy occurrence of antimicrobial-resistant pathogens in the freshwater system. Similar to the present study, multiple antibiotic resistance profiles have been detected among the enterobacteriaceae isolated from farmed catfish (Sarter et al. 2007).

The present study showed that all the recovered bacterial species were susceptible towards three antibiotics such

Fig. 3 A graph showing the percentage of antimicrobial resistance of goldfish isolates towards the 46 antibiotics



as ciprofloxacin, levofloxacin and perfloracin belonging to quinolone class. This suggests the possible application of those antibiotics for the immediate recovery of infected fish. However, the usage of antibiotics should be reduced and limited to urgent situations so as to minimize the development of AMR. The susceptibility of fish pathogens towards ciprofloxacin has been noticed in several cases as studied by Akinbowale et al. (2006), Kumar et al. (2015) and Preena et al. (2019a, b). Among the quinolone class tested, nalidixic acid was the ineffective one for almost 62% of the tested isolates. As stated by Akinbowale et al. (2006), the resistance of cultured fish pathogens towards nalidixic acid of quinolone class is common in aquaculture. Besides, ampicillin of beta-lactam group and vancomycin of glycopeptide group were completely ineffective against all the recovered bacteria on antimicrobial susceptibility tests. Likewise bacitracin of glycopeptide class was also ineffective against 85% of the studied isolates. This implied the severe resistance developed in the isolates against the glycopeptide class of antibiotics. Osman et al. (2016) had noticed severe glycopeptide resistance among *Enterococcus* isolates from Nile Tilapia. Around 77% and 70% of isolates possessed resistance against all tested antibiotics of tetracyclines and macrolides, respectively. Among the sulphonamides, sulphadiazine was the most resistant group for more than 50% of isolates. Aminoglycosides were also found to be ineffective for the treatment of around 45% of isolates. Another important thing to be pointed is the resistance of isolates towards the new generation cephalosporins. Besides developing resistance against beta lactams (60%), more than 59% of isolates showed AMR against first-generation cephalosporins with cephalothin (77%) ranked the topmost resistant one. Even though other new-generation antibiotics belonging to 2nd, 3rd and 4th cephalosporins and carbapenem groups are effective for the majority of the tested bacteria, around 35% of the isolates exhibited resistance against those classes. This could raise serious threats in aquaculture for successful management. As Liakopoulos et al. (2016) suggested, the prime reason for the resistance of enterobacteriaceae family members towards β -lactam antibiotics are the production of β -lactamases. The extended pectrum β -lactamase (ESBL) genes are reported to exist most often in mobile genetic elements for the inactivation of cephalosporins, which again threatens the context of HGT (Adelowo et al. 2018). Amoxicillin, aztreonam, cefazolin, nitrofurantoin, colistin and rifampicin are the other resistant antibiotics for more than 60% of isolates. While, chloramphenicol, trimethoprim, co-trimoxazole, norfloxacin, enrofloxacin, carbapenems and ceftriaxone are the least resistant antibiotics for around 70% of isolates. As a whole, amidst the 15 classes of antibiotics tested, glycopeptides are the most resistant groups followed by tetracyclines, macrolides, β lactams, first generation cephalosporins, and nitrofurans for most of

the isolates studied. While, quinolones, carbapenems and phenicol groups of antibiotics are the effective antibiotics for the same.

Antibiogram profile of enterobacteriaceae after plasmid curing

For determining the exact location of AMR genes in bacteria and to predict whether the antimicrobial resistance is of plasmid mediated or chromosome mediated, plasmids of the resistant isolates have been extracted. Ninety-two percent of the recovered isolates were found to carry plasmids, which highlight the heavy occurrence of plasmids in the tested bacteria. All the tested isolates except *K. cryocrescens* harbored plasmid with a molecular weight of > 10 kb and were subjected to plasmid curing. After testing a range of 0.1 to 1 mg/ml concentration of acridine orange for all the resistant isolates, 0.2 mg/ml of acridine orange was optimized for plasmid curing, which has been followed in several studies (Letchumanan et al. 2015b). After curing, plasmid extraction was repeated for the confirmation of elimination of plasmids and found to be lost from all the cured isolates as visualized using an agarose gel. Once the plasmid elimination has assured, those resistant isolates were again subjected to antibiotic susceptibility tests. Altered resistance phenotype was noticed after plasmid curing and the antibiogram profile varied with the different isolates tested. The antibiogram profile and MAR index of the recovered isolates from goldfish before and after plasmid curing are given in Table 3. Since *K. cryocrescens* hardly harbored plasmid, the antibiotic resistance observed is of chromosome mediated. The least antimicrobial resistant isolate, *C. freundii* was found to be susceptible towards all the antibiotics after plasmid curing, indicating that their all AMR genes are of plasmid mediated. This could raise the probability of a rapid spread of AMR throughout the system.

The MAR index of the most resistant isolate, *P. penneri* has been changed from 0.87 to 0.37 on testing antimicrobial susceptibility after plasmid curing. This implied that most of the AMR genes reside in mobile genetic elements, while tetracycline, erythromycin, nitrofurantoin, glycopeptides and some beta lactams resistance genes locate in chromosome. Likewise, *P. hauseri* also exhibited a similar kind of variation in MAR index from 0.65 to 0.32 by maintaining resistance to beta lactams, tetracyclines, erythromycin, glycopeptides, polymyxin and rifampicin after plasmid curing. In the case of *E. coli*, a drastic change in MAR index from 0.71 to 0.1 has been noticed, which suggested that almost many of the significant ARG have been lost on plasmid curing except tetracycline, rifampicin, vancomycin, polymyxin and ampicillin resistant genes. Similarly, the AMR of *K. aerogenes* towards most of the antibiotics had also lost, with the exception of some beta lactams, vancomycin, rifampicin,

Table 3 Chart showing the antibiogram profile and MAR index of the goldfish isolates before and after plasmid curing

Species	Genbank accession no.	Antibiogram phenotype before curing	Antibiogram phenotype profile before plasmid curing	MAR index before plasmid curing	Antibiogram phenotype profile after plasmid curing	MAR index after plasmid curing
<i>Proteus hauseri</i>	MT263016	Amikacin/ampicillin/amoxicillin/azithromycin/aztreonam/cefalexin/cefazolin/cefepime/cefixime-clavulanic acid/cefoperazone/cefotaxime/cefoxitin/ceftazidime/cefuroxime/cephalothin/colistin/bacitracin/erythromycin/furazolidone/gentamycin/halidixic acid nitrofurantoin/oxytetracycline/, piperacillin/piperacillin-tazobactam, polymyxin/rifampicin/streptomycin/sulphadiazine/vancomycin	Amikacin/ampicillin/amoxicillin/amoxycylav/aztreonam/bacitracin/cefalexin/cefazolin/cefixime-clavulanic acid/cefepime/cefoperazone/cefotaxime/cefotaxime-clavulanic acid/cefoxitin/ceftazidime/ceftazidime-clavulanic acid/ceftriaxone/cefuroxime/cephalothin/chloramphenicol/colistin/co-trimazole/doripenem/enrofloxacin/erythromycin/furazolidone/imipenem/nalidixic acid/nitrofurantoin/norfloxacin/oxytetracycline/piperacillin/piperacillin-tazobactam/polymyxin-B/rifampicin/streptomycin/sulphadiazine/tobramycin/trimethoprim/vancomycin	0.65	Ampicillin/amoxicillin/bacitracin/cephalexin/cefazolin/cefuroxime/cephalothin/colistin/erythromycin/furazolidone/nitrofurantoin/oxytetracycline piperacillin/piperacillin-tazobactam/polymyxin-B/rifampicin/trimethoprim/vancomycin	0.32
<i>Proteus penneri</i>	MT263017	Amikacin/ampicillin/amoxicillin/amoxycylav/aztreonam/bacitracin/cefalexin/cefazolin/cefixime-clavulanic acid/cefepime/cefoperazone/cefotaxime/cefotaxime-clavulanic acid/cefoxitin/ceftazidime/ceftazidime-clavulanic acid/ceftriaxone/cefuroxime/cephalothin/chloramphenicol/colistin/co-trimazole/doripenem/enrofloxacin/erythromycin/furazolidone/imipenem/nalidixic acid/nitrofurantoin/norfloxacin/oxytetracycline/piperacillin/piperacillin-tazobactam/polymyxin-B/rifampicin/streptomycin/sulphadiazine/tobramycin/trimethoprim/vancomycin	Amikacin/ampicillin/amoxicillin/amoxycylav/aztreonam/bacitracin/cefalexin/cefazolin/cefixime-clavulanic acid/cefepime/cefoperazone/cefotaxime/cefotaxime-clavulanic acid/cefoxitin/ceftazidime/ceftazidime-clavulanic acid/ceftriaxone/cefuroxime/cephalothin/chloramphenicol/colistin/co-trimazole/doripenem/enrofloxacin/erythromycin/furazolidone/imipenem/nalidixic acid/nitrofurantoin/norfloxacin/oxytetracycline/piperacillin/piperacillin-tazobactam/polymyxin-B/rifampicin/streptomycin/sulphadiazine/tobramycin/trimethoprim/vancomycin	0.87	Ampicillin/amoxicillin/bacitracin/cefalexin/cefazolin/cephalothin/colistin/erythromycin/furazolidone/nitrofurantoin/oxytetracycline piperacillin/piperacillin-tazobactam/polymyxin-B/rifampicin/trimethoprim/vancomycin	0.37
<i>Serratia marcescens</i>	MT263018	Amikacin/ampicillin/amoxicillin/bacitracin/cefalexin/cefazolin/cefoxitin/cefuroxime/cephalothin/colistin/furazolidone/nalidixic acid/nitrofurantoin/oxytetracycline/polymyxin-B/rifampicin/vancomycin	Ampicillin/amoxicillin/bacitracin/cefalexin/cefazolin/cefoxitin/cefuroxime/cephalothin/colistin/furazolidone/nalidixic acid/nitrofurantoin/oxytetracycline/polymyxin-B/rifampicin/vancomycin	0.35	Ampicillin/bacitracin/cephalexin/cefazolin/cefoxitin/cephalothin/polymyxin-B/rifampicin/vancomycin	0.19
<i>Citrobacter freundii</i>	MT263019	Ampicillin/amoxicillin/azithromycin/amoxycylav/bacitracin/erythromycin/nitrofurantoin/norfloxacin/polymyxin-B/vancomycin	Ampicillin/amoxicillin/azithromycin/amoxycylav/bacitracin/erythromycin/nitrofurantoin/norfloxacin/polymyxin-B/vancomycin	0.22	Nil	0
<i>Klebsiella aerogenes</i>	MT263020	Amoxycylav/amikacin/ampicillin/amoxicillin/azithromycin/bacitracin/cefalexin/cefazolin/cefepime/cefixime-clavulanic acid/cefoperazone/ceftazidime/ceftazidime-clavulanic acid/cefotaxime/cefuroxime/cephalothin/colistin/erythromycin/gentamycin/meropenem/halidixic acid/nitrofurantoin/oxytetracycline/piperacillin/piperacillin-tazobactam/polymyxin/rifampicin/streptomycin/sulphadiazine/tobramycin/vancomycin	Amoxycylav/amikacin/ampicillin/amoxicillin/azithromycin/bacitracin/cefalexin/cefazolin/cefepime/cefixime-clavulanic acid/cefoperazone/ceftazidime/ceftazidime-clavulanic acid/cefotaxime/cefuroxime/cephalothin/colistin/erythromycin/gentamycin/meropenem/halidixic acid/nitrofurantoin/oxytetracycline/piperacillin/piperacillin-tazobactam/polymyxin/rifampicin/streptomycin/sulphadiazine/tobramycin/vancomycin	0.67	Ampicillin/amoxicillin/bacitracin/ceftazidime/Ceftazidime-clavulanic acid/cephalothin colistin/piperacillin-tazobactam/rifampicin/tobramycin/vancomycin	0.24

Table 3 (continued)

Species	Genbank accession no.	Antibiogram phenotype profile before curing	Antibiogram phenotype profile before plasmid curing	MAR index before plasmid curing	Antibiogram phenotype profile after plasmid curing	MAR index after plasmid curing
<i>Plesiomonas shigelloides</i>	MT263021	Amikacin/ampicillin/amoxicillin/amoxycylav/aztreonam/bacitracin/cefepime/cefoperazone/erythromycin/nalidixic acid/streptomycin/tobramycin/vancomycin	Amikacin/ampicillin/amoxicillin/amoxycylav/aztreonam/bacitracin/cefepime/cefoperazone/erythromycin/nalidixic acid/streptomycin/tobramycin/vancomycin	0.28	Ampicillin/amoxicillin/bacitracin/vancomycin	0.08
<i>Edwardsiella tarda</i>	MT263022	Ampicillin/amoxicillin/azithromycin/aztreonam/bacitracin/cefoperazone/cefazidime/cephalothin/colistin/gentamycin/oxytetracycline/rifampicin/streptomycin/sulphadiazine/tobramycin/vancomycin	Ampicillin/amoxicillin/azithromycin/aztreonam/bacitracin/cefoperazone/cefazidime/cephalothin/colistin/gentamycin/oxytetracycline/rifampicin/streptomycin/sulphadiazine/tobramycin/vancomycin	0.35	Rifampicin/vancomycin	0.04
<i>Enterobacter cancerogenus</i>	MT263023	Ampicillin/amoxicillin/azithromycin/aztreonam/bacitracin/cefazolin/cefoperazone/cefotaxime/cephalothin/chloramphenicol/co-trimoxazole/imipenem/meropenem/nitrofurantoin/oxytetracycline/sulphadiazine/vancomycin	Ampicillin/amoxicillin/azithromycin/aztreonam/bacitracin/cefazolin/cefoperazone/cefotaxime/cephalothin/chloramphenicol/co-trimoxazole/imipenem/meropenem/nitrofurantoin/oxytetracycline/sulphadiazine/vancomycin	0.35	Ampicillin/amoxicillin/bacitracin/cefazolin/cephalothin/vancomycin	0.13
<i>Enterobacter ludwigii</i>	MT263024	Ampicillin/amoxicillin/azithromycin/aztreonam/bacitracin/cefazidime, cefazidime-clavulanic acid/colistin/furazolidone/nalidixic acid/nitrofurantoin/polymyxin-B/vancomycin	Ampicillin/amoxicillin/azithromycin/aztreonam/bacitracin/cefazidime, cefazidime-clavulanic acid/colistin/furazolidone/nalidixic acid/nitrofurantoin/polymyxin-B/vancomycin	0.28	Aztreonam/cefazidime/cefazidime-clavulanic acid/colistin/nalidixic acid	0.1
<i>Enterobacter cloacae</i>	MT263025	Ampicillin/azithromycin/aztreonam/cefazolin/cephalothin/erythromycin/gentamycin/rifampicin/tetracycline/vancomycin	Ampicillin/azithromycin/aztreonam/cefazolin/cephalothin/erythromycin/gentamycin/rifampicin/tetracycline/vancomycin	0.22	Rifampicin, vancomycin, cephalothin, ampicillin, erythromycin, cefazolin	0.13
<i>Escherichia coli</i>	MT263026	Amikacin/ampicillin/azithromycin/cefalexin/cefazolin/cefixime-clavulanic acid/cefoperazone/cefotaxime/cefotaxime-clavulanic acid/cefotaxime/cefotaxime-clavulanic acid/cefotaxime/cefotaxime-clavulanic acid/cefuroxime/cephalothin/colistin/co-trimoxazole/chloramphenicol/doripenem/enrofloxacin/erythromycin/meropenem/nalidixic acid/norfloxacin/oxytetracycline/piperacillin/piperacillin/tazobactam polymyxin-B/gentamycin/rifampicin/streptomycin/sulphadiazine/tobramycin/vancomycin	Amikacin/ampicillin/azithromycin/cefalexin/cefazolin/cefixime-clavulanic acid/cefoperazone/cefotaxime/cefotaxime-clavulanic acid/cefotaxime/cefotaxime-clavulanic acid/cefuroxime/cephalothin/colistin/co-trimoxazole/chloramphenicol/doripenem/enrofloxacin/erythromycin/meropenem/nalidixic acid/norfloxacin/oxytetracycline/piperacillin/piperacillin/tazobactam polymyxin-B/gentamycin/rifampicin/streptomycin/sulphadiazine/tobramycin/vancomycin	0.71	Ampicillin/oxytetracycline/polymyxin/rifampicin/vancomycin	0.1
<i>Providencia vermicola</i>	MT263027	Ampicillin/amoxicillin/azithromycin/bacitracin/cefalexin/cefazolin/cefepime/cefixime-clavulanic acid/cefoperazone/cefazidime/cefazidime-clavulanic acid/cefotaxime/cefuroxime/cephalothin/colistin/erythromycin/furazolidone/gentamycin/nalidixic acid/nitrofurantoin/oxytetracycline/polymyxin-B/piperacillin/piperacillin/tazobactam/rifampicin/sulphadiazine/trimethoprim/vancomycin	Ampicillin/amoxicillin/azithromycin/bacitracin/cefalexin/cefazolin/cefepime/cefixime-clavulanic acid/cefoperazone/cefazidime/cefazidime-clavulanic acid/cefotaxime/cefuroxime/cephalothin/colistin/erythromycin/furazolidone/gentamycin/nalidixic acid/nitrofurantoin/oxytetracycline/polymyxin-B/piperacillin/piperacillin/tazobactam/rifampicin/sulphadiazine/trimethoprim/vancomycin	0.61	Ampicillin/bacitracin/cefalexin/colistin/erythromycin/oxytetracycline/poly myxin-B/vancomycin	0.17

Table 3 (continued)

Species	Genbank accession no.	Antibiogram phenotype profile before plasmid curing	MAR index before plasmid curing	Antibiogram phenotype profile after plasmid curing	MAR index after plasmid curing
<i>Kluyvera cryocrescens</i>	MT2630128	Ampicillin/amoxicillin/amoxycylav/azithromycin/aztreonam/bacitracin/cefazolin/cephalothin/erythromycin/nitrofurantoin/oxytetracycline/rifampicin/vancomycin	0.28	Ampicillin/amoxicillin/amoxycylav/azithromycin/aztreonam/bacitracin/cefazolin/cephalothin/erythromycin/nitrofurantoin/oxytetracycline/rifampicin/vancomycin	0.28

bacitracin resistance on plasmid curing, with a shift of MAR index from 0.67 to 0.24. *P. vermicola* had also lost many of ARG except those coding for vancomycin, tetracycline, ampicillin, rifampicin, bacitracin resistance with the lower MAR index of 0.17. While *E. tarda* retained only vancomycin and rifampicin resistance (MAR index = 0.04) after plasmid curing indicating almost all the ARG are of plasmid borne.

Like all other isolates, vancomycin, bacitracin, polymyxin, rifampicin and some beta lactam resistance property could be retained among *S. marcescens* and *P. shigeloides* after plasmid curing with the MAR index of 0.19 and 0.08, respectively. All the two species of *Enterobacter* such as *E. cancerogenus* and *E. cloacae* exhibited a similar kind of antibiogram profile after plasmid elimination, having the MAR index of 0.13. However, *E. ludwigii* had retained resistance only towards nalidixic acid, aztreonam, ceftazidime/clavulanic acid and colistin with very low MAR index of 0.1. Meanwhile, vancomycin, bacitracin, ampicillin and rifampicin resistance were retained among the majority of the strains after plasmid curing, highlighting their existence in the chromosome. The lack of evident resistance in most of the isolates towards new generation cephalosporins, carbapenems, tetracycline, aminoglycosides, macrolides and quinolones after plasmid curing signifies the location of those resistant genes is in plasmid. This highlights the increased possibility of heavy AMR dissemination in aquaculture through conjugation and transformation. This is in close agreement with the recovery of plasmids from 70% of ESBL positive enterobacteriaceae in aquaculture ecosystem (Adelowo et al. 2018). The mean MAR index value of tested isolates after plasmid curing was found to be 0.16 which made a great difference than the mean value (0.45) before plasmid elimination. This again highlights the intensive change of AMR among the isolates after plasmid curing and the severity of plasmid-mediated AMR. Lunn et al. (2010) already stated that plasmid-mediated transferable AMR determinants could be observed mainly among enterobacteriaceae than vibrios and aeromonas. Thus, the present study revealed that the studied infected fish are the huge reservoirs of diverse ARGs in the aquaculture system. As a remedy, the initiative of one health approach, different management measures, alternative strategies and timely detection methods to unravel the complexity of AMR in aquaculture were reviewed in detail by various researchers (Preena et al. 2020a, b).

Conclusion

To the best of our knowledge, this is the first comprehensive report mentioning the prevalence of antibiotic resistance in enterobacteriaceae family members recovered from

the diseased goldfish before and after plasmid curing. This study emphasizes the potential of the pathogens to resist against the diverse classes of drugs administered in aquaculture. Resistance of pathogens towards new generation cephalosporins could raise major challenges. The occurrence of plasmids in 92% of studied enterobacteriaceae isolates and their plasmid-mediated AMR pointed out the severe spread of AMR in aquaculture ecosystem. In this scenario, effective management measures have to be undertaken to curb the spread of AMR in aquaculture.

Author's contributions Following contributions had been done by the authors: Conceived of or designed study: PGP and TRS; Performed research: PGP, AD, SRN, ARS and RRN. Data analysis and interpretation: PGP, AD and TRS. Manuscript draft: PGP and TRS. All authors read and approved the final manuscript.

Funding The authors thank Director, ICAR-National Bureau of Fish Genetic Resources, Lucknow for the support and ICAR National Fellow project (F. No Agri.edn.27/2/2015) for funding and one of the authors, Dr. Preena P. G thanks Department of Science and Technology (SERB) for providing National Postdoctoral Research fellowship (PDF/2017/000378).

Availability of data and material Not applicable.

Code availability Not applicable.

Compliance with ethical standards

Conflict of Interest The authors have no conflict of interest to declare.

Ethics approval All the experimental challenge procedures in this study (Proposal number: NBFGR/IAEC/2019/0014) were evaluated and approved by Institute Animal ethics Committee (IAEC) of ICAR National Bureau of Fish Genetic Resources (NBFGR) (CPCSEA Registration No: 909/GO/Re/S/05/CPCSEA dated 09.09.2005 and CPCSEA Ref file No. 25/111/2014-CPCSEA dated 05th December 2018) vide approval Number G/IAEC/2019/1 dated 04th October 2019.

Consent to participate Not applicable.

Consent for publication Not applicable.

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